

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		frequency of HbA = 0.96 (1) frequency of HbS = 0.04 (1) frequency of heterozygotes = 0.0768/ 0.08/ 0.077 (1) ecf from above % frequency of heterozygotes = 7.68 / 8.0/ 7.7 % (1) ecf from frequency		4		4	4	
	(b)			Any five × (1) from: A. {HbS HbS/ or description} are not likely to survive (1) <u>African-Americans</u> B. no (selective) advantage to being heterozygote (1) C. Disadvantage to being {HbS HbA / heterozygotes/ sickle trait} / {HbS HbA / heterozygotes/ sickle trait} are selected against (1) D. there is an advantage to being {HbA HbA/ homozygous HbA} (1) <u>Africa</u> E. HbA HbA more likely to die of malaria / selected against. (1) F. {Heterozygotes/ HbA HbS/ having one HbS } (selective) advantage/ {Heterozygotes/ HbA HbS/ having one HbS } are selected for / or description of (1)			5	5		
	(c)	(i)		Any two × (1) from: {Nucleotide /base/DNA} { <u>sequence/ code</u> } (1) Location of genes (1) Accept alleles on chromosomes (1) complete DNA sequence = 2 marks	2			2		